

Shaan Chopra

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Education

University of Washington (UW)

PHD IN COMPUTER SCIENCE & ENGINEERING

Seattle, WA, USA

Sep 2021 - Present

Advised by James Fogarty & Sean Munson | CGPA: 3.95 | **Courses:** HCI, AI, Human-AI Interaction, Ethics, Privacy, Critical Technology Practice

University of Maryland - College Park (UMD)

MASTER OF SCIENCE IN HUMAN COMPUTER INTERACTION

College Park, MD, USA

Aug 2019 - May 2021

Advised by Eun Kyoung Choe | CGPA: 4.0 | **Courses:** Health Informatics, Data Visualization, UX Research, Interaction Design, Makerspaces

Indraprastha Institute of Information Technology, Delhi (IIIT-D)

B.TECH. IN COMPUTER SCIENCE & ENGINEERING

New Delhi, India

Aug 2015 - May 2019

CGPA: 8.7/10 | **Courses:** Data Structures, Algorithms, OOP, Databases, OS, Biometrics, Human-Centered Design, Wearable Applications

Skills

Research & Analysis

Human-Centered Design • AI Applications for Health & Wellbeing • Personal Informatics & Technology Design • User Experience • Participatory Design • Study Design • Qualitative Research & Analysis • Mixed-Methods • Data Visualization • Community-Engaged Research

Languages, Tools, and Technologies

Python • C/C++ • Java • Ruby • MATLAB • HTML • CSS • JavaScript • Flask • Android Studio • Keras • SQL • NoSQL • MongoDB • AngularJS • Spring • D3.js • Tableau • $\LaTeX$ • Tinkercad • Figma • InVision • Balsamiq • Marvel • Sketch • Photoshop • Qualtrics • UserZoom • NVivo • Dedoose

Selected Peer-Reviewed Publications

- P13 **S. Chopra***, A. Okeson*, Y. S. Sefidgar, J. Fogarty, S. A. Munson: **Bayesian Analysis of Self-Trackd Health Data: A Technology Probe Study Examining Opportunities in Integration and Reflection.** Interactive Health'26. 🔗
- P12 **S. Chopra**, K. Juarez, J. Fogarty, S. A. Munson: **Engagements with Generative AI and Personal Health Informatics: Opportunities for Planning, Tracking, Reflecting, and Acting around Personal Health Data.** IMWUT'25. 🔗
- P11 S. A. Kraft, **S. Chopra**, M. Duran, J. A. Rojina, A. Beretta, K. I. López, R. Javan, B. S. Wilfond, M. Rosenfeld, J. Fogarty, L. K. Ko: **Community Perspectives on Artificial Intelligence-Enabled Mobile Health Tools: A Focus Group Study of Hispanic and Latinx Community Members.** Journal of Medical Internet Research. 🔗
- P10 A. Pradhan, S. Erete, **S. Chopra**, P. Upadhyay, O. Sule, A. Lazar: **No, not that voice again!: Engaging Older Adults in Design of Anthropomorphic Voice Assistants.** CSCW'25. 🔗
- P9 **S. Chopra**, L. Orii, K. Juarez, N. Tieanklin, J. Fogarty, S. A. Munson: **Menopause Legacies: Designing to Record and Share Experiences of Menopause Across Generations.** CSCW'24. 🔗
- P8 **S. Chopra**, J. Carroll, J. Pater: **Providing Context to the “Unknown”: Patient and Provider Reflections on Connecting Personal Tracking, Patient-Reported Insights, and EHR Data within a Post-COVID Clinic.** CSCW'24. 🔗
- P7 J. Pater, **S. Chopra**, J. Zaccour, J. Carroll, F. F. Nova, T. Toscos, S. Guha, F. L. Chang: **Charting the COVID Long Haul Experience – A Longitudinal Exploration of Symptoms, Activity, and Clinical Adherence.** CHI'24. 🔗
- P6 Y. S. Sefidgar, C. L. Castillo, **S. Chopra**, L. Jiang, T. Jones, A. Mittal, H. Ryu, J. Schroeder, A. Cole, N. Murinova, S. A. Munson, J. Fogarty: **MigraineTracker: Examining Patient Experiences with Goal-Directed Self-Tracking for a Chronic Health Condition.** CHI'24. 🔗
- P5 **S. Chopra**, A. Beretta, M. Duran, K. I. López, J. A. Rojina, L. K. Ko, M. Rosenfeld, B. S. Wilfond, J. Fogarty, S. A. Kraft: **Developing a Resource for Supporting Community-Based Health Research: Towards Considerations for Advancing Equity in Mobile Health Technology.** CSCW'24 Poster. 🔗
- P4 **S. Chopra**, R. Zehrung, T. Shanmugam, E. K. Choe: **Living with Uncertainty and Stigma: Self-Experimentation and Support-Seeking around Polycystic Ovary Syndrome.** CHI'21. 🔗
- P3 **S. Chopra**, E. Dixon, K. Ganesh, A. Pradhan, M. Radnofsky, A. Lazar: **Designing for and with People with Dementia using a Human Rights-Based Approach.** CHI'21 Case study. 🔗
- P2 A. Tuli, **S. Chopra**, P. Singh, N. Kumar: **Menstrual (Im)Mobilities and Safe Spaces.** CHI'20. 🔗
- P1 A. Tuli, **S. Chopra**, N. Kumar, P. Singh: **Learning from & with Menstrupedia: Towards Menstrual Health Education in India.** CSCW'18. 🔗

Industry Experience

Merck Sharp & Dohme (MSD)

Remote (West Point, PA)

RESEARCH INTERN (UX & EPIDEMIOLOGY)

May 2025 - Aug 2025

- Worked with Real World Evidence Capabilities and Analytics Team under Biostatistics and Research Decision Sciences (BARDS).
- Conducted **user research** and **usability testing** for multiple projects involving real world data platforms and partnerships.
- Informed design of different capabilities, including potential opportunities to incorporate generative AI, to support decision-making around and analysis of real world evidence health data.

Institute of Translational Health Sciences (ITHS) - UW

Seattle, WA

RESEARCH INTERN

Jun 2024 - Sep 2024

- Worked with the ITHS Hub Liaison & Network Capacity Team on a translational science project to improve participant diversity and enrolment from underserved populations in clinical trials, with a salient focus on use of remote technologies.
- Conducted **interviews & focus groups** with clinical investigators and research coordinators.
- Created a bundle of interventions to share with (i) clinical researcher teams and (ii) institutional research support services across UW Medicine, Fred Hutch Cancer Research Center, and Seattle Children's Hospital.
- Investigated implementation pathways for using identified interventions within real world constraints (e.g., technology access barriers, social determinants and health disparities affecting different communities).

Population Health Initiative (PHI) - UW

Seattle, WA

SOCIAL ENTREPRENEURSHIP FELLOW

Jun 2023 - Aug 2023

- Worked as a part of the PHI to **understand the market need & develop a strategy for adoption** of the new, culturally-responsive child developmental screening tool being developed by the project team (Julie Kientz, Kendra Liljenquist).
- Conducted **customer discovery & market research**, interviewing various stakeholders.
- Provided recommendations for: cost-measuring, policy, legal and regulatory issues, scaling, financial models, and tool sustainability 📄.

Parkview Mirro Center for Research and Innovation

Remote (Fort Wayne, IN)

RESEARCH INTERN

Jun 2022 - Sep 2022

- Examined data needs & communication practices between clinicians & long covid patients.
- Conducted **interviews & focus groups** with medical providers.
- Used **qualitative & quantitative methods** and **data visualization** to analyze **health tracking, survey, interview & health record data**.
- **Designed dashboards & digital formats** for visualizing patient personal informatics data; dashboards were integrated into Parkview Health's clinical infrastructure in 2022 and are still used by clinicians for creating personalized plans-of-care for patients.
- Work resulted in 2 peer-reviewed publications P7 📄 and P8 📄.

Selected Research Projects

Examining Use of Generative AI for Reflecting & Making Sense of Personal Health Informatics Data

- Examined potential of Generative AI to support people in gaining personalized & actionable insights from their self-tracked health data.
- Used **qualitative & quantitative research methods** to study how people prompt & perceive health data insights by Generative AI.
- Analyzed data using **hybrid deductive & inductive thematic analysis** to identify needs and opportunities for generative AI to scaffold support across self-tracking stages of planning, reflecting, and acting on health data insights.
- Work resulted in a peer-reviewed publication P12 📄 and directly enabled a new National Science Foundation award developing human-centered AI in personal health informatics (NSF IIS-2551299, \$900k).

Examining Community-Responsive AI-enabled Mobile Health Tools for Hispanic & Latino Populations

- Collaborated with research ethicists, pediatric pulmonologists & public health researchers Seattle Children's Hospital to examine Hispanic & Latinx community perspectives of AI-enabled mHealth tools for monitoring pediatric asthma.
- Conducted **focus group sessions** in rural & urban Washington to explore health condition & population-specific considerations for clinical utility, usability, burden, privacy & data sharing in AI-enabled mHealth tools.
- Led the development of a translational resource for supporting cultural-responsiveness in mHealth and AI/ML research in health P5 📄
- Work resulted in a peer-reviewed publication P11 📄.

Designing to Record & Share Menopause Experience Across Generations

- Studied practices for recording & sharing menopause experiences across generations and within families.
- Created **design sketches & probes** for recording & sharing menopause legacies to use in **remote co-design** sessions.
- Work resulted in peer-reviewed publication P9 📄.

Tracking and Managing Stigmatized Enigmatic Health Conditions

- Examined experiences of people with enigmatic conditions (e.g., PCOS) with the aim to support their tracking and collaboration needs.
- Conducted **in-depth interviews, participatory design & research**, and **data analysis** on an online social support platform.
- Work resulted in a peer-reviewed publication P4 📄 and master's thesis 📄.

Design Methods for Dementia

- Collaborated with a dementia rights advocate and proposed **user-centered design methods** for designing for and with people with cognitive disabilities, particularly people with dementia.
- Designed and conducted workshops teaching aspiring UX designers the proposed method.
- Published a case study documenting our proposed method P3 📄.

Designing Voice-Assistant Technologies for Older Adults

- Collaborated on a project to understand voice assistant use by older adults.
- Assisted with designing the study, including conceptualization and material creation for the **participatory design** sessions.
- Conducted **remote co-design sessions** with older adults.
- Work resulted in a peer-reviewed publication P10 📄.

Awards & Achievements

- 2024 **Best Paper Award**, Received at ACM CHI 2024
- 2023 **Grace Hopper Conference Scholarship**, Received UW CSE funding to attend GHC 2023
- 2022 **TAPIA Conference Scholarship**, Received UW CSE funding to attend TAPIA Conference 2022
- 2021 **Wissner-Slivka Endowed Graduate Fellowship**, UW CSE 1st year fellowship
- 2020 **College of Information Studies Alumni Chapter Scholarship**, Exceptional UMD iSchool grad student award
- 2020 **Grace Hopper Conference Scholarship**, Received UMD iSchool scholarship to attend GHC 2020
- 2018 **ACM-Women Scholarship**, Received travel scholarship to attend CSCW 2018
- 2018 **Google IWD Scholarship**, Received travel scholarship to attend IWD Celebrations at Google, Bangalore
- 2018 **Dean's List for Academic Excellence**, 9.5+ CGPA in third year of undergraduate degree at IIIT-D

Teaching Experience

- CSE 510 **Human Computer Interaction**, with James Fogarty at UW *Spr'26, Fa'23*
- CSE 442 **Data Visualization**, with Jeffrey Heer at UW *Win'26, Fa'25, Fa'24*
- CSE 512 **Data Visualization**, with Jeffrey Heer at UW *Spr'25*
- CSE 440 **Human Computer Interaction**, with Scott Ichikawa at UW *Spr'23*
- INST 462 **Data Visualization**, with Babaki Fotouhi at UMD *Spr'21*
- INST 201 **Information Science**, with M. R. Sauter at UMD *Spr'20, Fa'20*
- INST 362 **User-Centered Design**, with Rachael Bradley-Montgomery at UMD *Fa'19*
- HSS 2xx **Digital Ethnography**, with Shriram Venkatraman at IIIT-D *Fa'17*

Academic Service

- 2026 **Web Co-Chair**, ACM Interactive Health Conference (IH)
- 2026 **Student Volunteer**, ACM Interactive Health Conference (IH)
- 2021-26 **Reviewer**, ACM Conference on Human Factors in Computing (CHI)
- 2025 **Reviewer**, ACM Transactions on Computing for Healthcare (HEALTH)
- 2025 **Reviewer**, ACM Transactions on Computer-Human Interaction (TOCHI)
- 2022-25 **Reviewer**, ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW)
- 2024-25 **Reviewer**, ACM Designing Interactive Systems (DIS)
- 2022 **Student Volunteer**, ACM Conference on Human Factors in Computing (CHI)
- 2021 **Web Co-Chair**, ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW)
- 2019 **Student Volunteer**, Conference on Information & Communication Technologies and Development (ICTD)